UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
ADVENT 2025 SEMESTER **EXAMINATION**

PROGRAM: Bachelor of Science in Data Science & Analytics

YEAR: 3 SEMESTER: 1

COURSE CODE: DSC3112

COURSE NAME: COGNITIVE COMPUTING

EXAMINATION TYPE: 100% PROJECT-BASED EXAM

PROJECT DURATION: December 2025

TIME ALLOWED: TWO WEEKS

Examination Instructions

1. The general Uganda Christian University examination guidelines, academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This is an individual project-based exam and should be executed in two weeks.
   i. Choose **ONE** of the mentioned scenarios for your cognitive agent.
   ii. Work through your selected scenario systematically across Parts **A**, **B**, and **C**.
   iii. Assessment of the project shall be based on **five** milestones. Each milestone shall be evaluated out of **20 marks.** (See milestone table).
   iv. To achieve all the milestones, students should submit their work as python notebooks with GitHub repository links under the students' personal accounts.
3. Students should prepare to make a presentation of their work on the day scheduled for this exam. This will be a physical presentation. Any candidate who fails to present during their presentation slot will not be given another opportunity.
4. Students will also be required to submit soft copies of their documentation and their system on the UCU moodle platform.
5. Any student who does not present his/her work on the scheduled examination day will be considered to have missed the exam.

Project Philosophy: Architecting a Cognitive Solution

This project serves as the practical application of your studies in Cognitive Computing. Having explored the theories that allow machines to mimic human-like intelligence, you are now tasked with architecting a functional system that embodies these principles. At the heart of your design, you must thoughtfully integrate the four pillars of cognitive systems to address a tangible Ugandan challenge:

1. Understand : Your system must move beyond surface-level data processing to interpret context, ambiguity, and meaning, especially from unstructured human language.
2. Reason : Once information is understood, your system must be able to use it. How does your agent use its knowledge base to reason its way to a sound answer?
3. Learn: How could your system incorporate new data or user feedback to refine its performance and knowledge over time?
4. Interact: How does your agent communicate its complex reasoning back to a user in a simple, clear, and context-aware manner?

Your project will be evaluated on the depth and coherence of your implementation across these four pillars. We are looking for a system that is not just technically functional, but thoughtfully engineered to be a true cognitive agent.

Project Scenarios: ***Choose ONE of the following scenarios for your cognitive agent.***

Scenario 1: Agricultural Advisory Agent

A small-scale farmer near Mukono observes unusual spots on her maize leaves. She submits a photo along with a query in mixed Luganda and English: "What is this on my crops? What should I do about it?".
Your agent is expected to perceive this multimodal input, understand the potential crop disease from both the text and image, and reason over a knowledge base of local crop data and weather patterns to provide clear, actionable advice.

- Key Cognitive Capabilities: Natural Language Processing, Image Classification, Knowledge Graphs.
- Relevant Datasets: PlantVillage Dataset, Ugandan weather data, local crop information.

Scenario 2: Community Health Diagnostic Assistant

A Village Health Team (VHT) member in a remote part of Gulu is assessing a patient with a fever and a rash. Using a simple interface, they describe the patient's symptoms in unstructured text. Your agent must perceive this description, understand the combination of symptoms in a medical context, and reason by comparing them against a knowledge base of common local illnesses to suggest potential diagnoses and recommend the nearest appropriate health facility.

- Key Cognitive Capabilities: Natural Language Processing, Machine Learning (Probabilistic Reasoning), Knowledge Representation.
- Relevant Datasets: Public disease symptom data, Uganda Ministry of Health data.

Scenario 3: Personalized Educational Recommender

A Computing student preparing for her final exams is struggling with a specific topic; Quantum computing. They submit a query to your platform: *"Explain the basics of quantum computing and show me how it could be relevant for solving problems in Uganda."*

Your agent must perceive the query, understand the core academic concept and its requested local context, and reason over its database of content to recommend a personalized learning path of videos, articles, and quizzes.

- Key Cognitive Capabilities: Recommendation Systems, Natural Language Processing, Knowledge Graphs.
- Relevant Datasets: open-source educational content, learning materials.

Scenario 4: Small Business Intelligence Analyst

The owner of a small business in Kampala wants to understand customer sentiment and identify market trends. Your system analyzes a stream of online customer reviews and social media posts. Your agent must perceive this informal text, understand the expressed sentiment and key topics (e.g., "late delivery," "good service," "demand in Nakawa"), and reason over this data to generate a concise intelligence report with predictive insights.

- Key Cognitive Capabilities: Sentiment Analysis, Topic Modeling (NLP), Predictive Modeling.
- Relevant Datasets: Local business news, public social media data, market data.

PART A: PROBLEM ANALYSIS & SYSTEM DESIGN [30 MARKS]

I. Task A1: Problem Analysis Document [10 Marks]

Author a comprehensive document that defines your chosen scenario within its Ugandan context. Include an analysis of key stakeholders, user personas, and a clear justification for the project's relevance to local development goals. Conduct a brief literature review of existing or similar solutions.

II. Task A2: System Architecture & Cognitive Design [10 Marks]

Design the complete architecture for your cognitive agent. This must include:

I. A system architecture diagram.

II. A detailed Cognitive Processing Pipeline diagram, illustrating how data flows through the system. You must explicitly map your components to the Understand, Reason, Learn and Interact pillars.

III. Task A3: Implementation Plan [10 Marks]

Develop a professional project plan, including a detailed work breakdown structure, a project timeline with clear milestones, and a risk assessment with mitigation strategies.

PART B: COGNITIVE SYSTEM IMPLEMENTATION [50 MARKS]

I. Task B1: Data Perception & Preparation [10 Marks]

This is the system's sensory input layer. Implement a robust data pipeline in a Python notebook to acquire, clean, and preprocess your chosen datasets.

This pipeline must demonstrate a thorough transformation of raw, unstructured data into a clean, standardized format ready for cognitive processing.

II. Task B2: The Understanding & Reasoning Engine [20 Marks]

You must implement and integrate at least two of the following capabilities to demonstrate a clear pathway from understanding information to reasoning with it:

I. Natural Language Processing: Implement techniques that capture deep meaning.

II. Knowledge Graph Construction: Extract entities and relationships from your data to build a structured knowledge base that your agent can query to perform reasoning tasks.

III. Machine Learning Models: Train and validate a model to perform classification, prediction, or another reasoning task that is central to your agent's function.

IV. Multimodal Understanding: If applicable to your scenario, develop a model to process non-text data (e.g., images, audio) and integrate its output with the results of your text analysis.

III. Task B3: The Interaction Layer [20 Marks]

Develop a working prototype of your agent with either a web-based or chatbot interface. This interface must allow a user to interact with the system and receive a response generated by your reasoning engine, demonstrating the complete cognitive cycle from input to output.

PART C: EVALUATION & PROFESSIONAL DELIVERABLES [20 MARKS]

I. Task C1: System Evaluation Report [10 Marks]

Conduct a rigorous evaluation of your prototype's performance. Analyze its strengths and weaknesses using both quantitative metrics and qualitative observations. Critically compare your cognitive approach to a simpler, baseline method (for example a keyword-based search). Conclude with recommendations for future improvements.

II. Task C2: Ethical & Impact Analysis [5 Marks]

Analyze the ethical implications and potential societal impact of your system. Your analysis must address:

A. Data Bias and Fairness: Investigate your training data for potential biases (e.g., linguistic, regional, demographic) and discuss their potential impact on your agent's fairness. Propose mitigation strategies.
B. Data Privacy and Contextual Appropriateness: Outline the measures required to protect user data and discuss whether your system's design is culturally and contextually appropriate for its intended Ugandan users.

III. Task C3: Professional Deliverables [5 Marks]

Submit a complete and professional project package, including a comprehensive final report, polished presentation slides, and a well-documented code repository with a clear user manual.

PROJECT MILESTONES & ASSESSMENT

S/N	Milestone Description	Maximum Marks
1	MILESTONE ONE System Blueprint & Data Pipeline: ● Problem document, Architecture diagrams, Data pipeline notebook.	20%
2	MILESTONE TWO The Understanding Engine: ● Notebooks for core cognitive models, Trained model files.	20%
3	MILESTONE THREE Interactive Prototype: ● Working web app/chatbot with integrated models, Source code.	20%
4	MILESTONE FOUR Evaluation & Ethical Review: ● System evaluation report, Ethics & impact assessment document.	20%
5	MILESTONE FIVE Final Documentation & Presentation: ● Final project report, Presentation slides, Code repository.	20%
	TOTAL MARKS	100%

~END OF EXAM QUESTIONS~

MERRY CHRISTMAS & HAPPY NEW YEAR!